

0 of 12 configurations escape DRBC review by averaging, and all 8 that would trigger it escape if they buy water from an existing system

Modeled daily cooling makeup compared with the DRBC review threshold of 100,000 gal/day, averaged over any 30 consecutive days.

IT load (MW)	Configuration	30-day avg, annual mean (gal/d) ¹	30-day avg, max (gal/d) ¹	Peak day p99 (gal/d) ¹	Peak day max (gal/d) ¹	Below trigger on 30-day avg ²	Days/yr, 30-day avg > 100,000 ²	Days/yr, single day > 100,000 ¹	Of those, flow < DOY p25 ³	Of those, flow < 7Q10 ³	Review if self-supplied ²	Review if purchased from existing system ⁵
50	Evaporative tower	751,000	751,000	751,000	751,000	No	365	365	24.9%	0%	Yes	No
50	Hybrid (calibrated)	20,400	212,000	352,000	664,000	No	24.8	27.1	23.9%	0%	Yes	No
50	Air-cooled chiller	0	0	0	0	Yes	0	0	n/a	n/a	No	No
100	Evaporative tower	1,500,000	1,500,000	1,500,000	1,500,000	No	365	365	24.9%	0%	Yes	No
100	Hybrid (calibrated)	40,700	424,000	705,000	1,330,000	No	56.8	35.8	24.1%	0%	Yes	No
100	Air-cooled chiller	0	0	0	0	Yes	0	0	n/a	n/a	No	No
200	Evaporative tower	3,010,000	3,010,000	3,010,000	3,010,000	No	365	365	24.9%	0%	Yes	No
200	Hybrid (calibrated)	81,400	849,000	1,410,000	2,660,000	No	88.4	41.3	24.5%	0%	Yes	No
200	Air-cooled chiller	0	0	0	0	Yes	0	0	n/a	n/a	No	No
400	Evaporative tower	6,010,000	6,010,000	6,010,000	6,010,000	No	365	365	24.9%	0%	Yes	No
400	Hybrid (calibrated)	163,000	1,700,000	2,820,000	5,310,000	No	105	44.6	25.3%	0%	Yes	No
400	Air-cooled chiller	0	0	0	0	Yes	0	0	n/a	n/a	No	No
n/a	Falls Twp (AWS Keystone), reported	135,000 ⁴	1,410,000 ¹	n/a	4,400,000 ⁴	No	n/a	n/a	n/a	n/a	Yes	No

No configuration in the grid falls in the averaging blind spot, defined as below the threshold on every 30-day average but above 100,000 gal on at least one day. Under a looser annual-mean test, 3 of 12 configurations would qualify. The Falls row is a reference and is not counted.

¹ Model-derived: PUE 1.2, 4 cycles of concentration, hybrid parameters from results/calibration.json primary, KTTN hourly wet-bulb 2005-2024 [noaa_isd, stull_2011, alduchov_1996]. Makeup is treated as gross withdrawal. 101 of 7,305 local days are missing, and 30-day windows require at least 27 valid days.

² A withdrawal is excluded from review when the daily average gross withdrawal during any 30 consecutive day period does not exceed 100,000 gallons [drbc_admin_manual] (DRBC Administrative Manual Part 1, Article 3 and 18 CFR 401.35(a)(2)-(3)). Days/yr, 30-day avg counts days whose trailing 30-day average exceeds 100,000 gal/d (the trigger test). Days/yr, single day counts days whose own makeup exceeds 100,000 gal and is not the trigger test. Both are divided by the number of valid years (valid days or valid windows / 365.25).

³ Delaware River at Trenton daily mean flow on days above 100,000 gal, compared with its 2005-2024 day-of-year 25th percentile and the full-record LP3 7Q10 of 1,795 cfs [usgs_nwis_01463500].

⁴ Reported: average cooling service water 135,000 gal/day [falls_levittown_2026] and peak 4.4 million gal/day [falls_levittown_2026, falls_herald_2026].

⁵ The thresholds apply to a project's own withdrawal, so a data center that buys water from an existing public or authority system is not itself reviewed [drbc_datacenters_2026, drbc_admin_manual]. Falls is supplied by the Morrisville Municipal Authority service-water system, whose allocation is reported as 243 million gal/day [falls_levittown_2026].